Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

[Course Code: 23ECSPE1AD]

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Comparator (Combinational)

Theory:

A 4-bit comparator is a combinational circuit used to compare two 4-bit binary numbers A and B. It checks whether A is greater than, less than or equal to B and produces the corresponding output. If A is greater than B, A_gt_B becomes HIGH; if A is less than B, A_lt_B becomes HIGH; and if both inputs are equal, A_et_B becomes HIGH. Only one output is HIGH at a time. It is commonly used in digital systems, processors and arithmetic logic units

Code:

comp_4bit.v

```
module comparator_4bit (
     input [3:0] A,
     input [3:0] B,
     output reg GT,
     output reg LT,
     output reg EQ
);

  always @(*) begin
     if (A > B)  begin
          GT = 1'b1;
          LT = 1'b0;
          EQ = 1'b0;
     end
     else if (A < B)  begin
          GT = 1'b0;
          LT = 1'b1;
          EQ = 1'b0;
     end
     else begin
          GT = 1'b0;
          LT = 1'b0;
          EQ = 1'b1;
     end
end

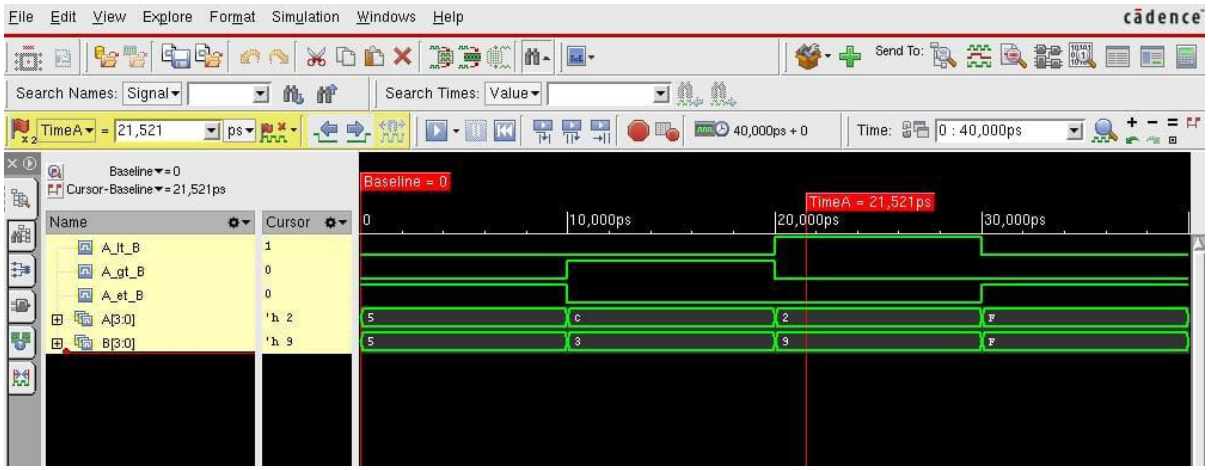
endmodule
```

Testbench:

comp_4bit_tb.v

```
1   `timescale 1ns/1ps
2   module comp_4bit_tb;
3
4      reg [3:0] A, B;
5      wire GT, LT, EQ;
6
7   comparator_4bit uut ( .A(A), .B(B), .GT(GT), .LT(LT), .EQ(EQ) );
8
9   initial  begin
10      A = 4'b0101; B = 4'b0011;  #10;  // A > B
11      A = 4'b0010; B = 4'b0110;  #10;  // A < B
12      A = 4'b1010; B = 4'b1010;  #10;  // A = B
13      A = 4'b1111; B = 4'b0001;  #10;  // A > B
14      A = 4'b0000; B = 4'b1111;  #10;  // A < B
15      $finish;
16  end
17
18  endmodule
19
20
```

Output Waveform:



Shift Register(Sequential)

Theory:

A shift register is a sequential logic circuit consisting of a group of flip-flops connected in series to store and transfer binary data. It shifts the stored data by one bit position either to the left or right for every clock pulse applied. Each flip-flop stores one bit of information, and the output of one flip-flop is connected to the input of the next. Shift registers are classified into four types: Serial-In Serial-Out (SISO), Serial-In Parallel-Out (SIPO), Parallel-In Serial-Out (PISO), and Parallel-In Parallel-Out (PIPO), based on how data is entered and retrieved. They are commonly used in temporary data storage, serial-to-parallel and parallel-to-serial data conversion, and data transfer in digital systems.

Code:

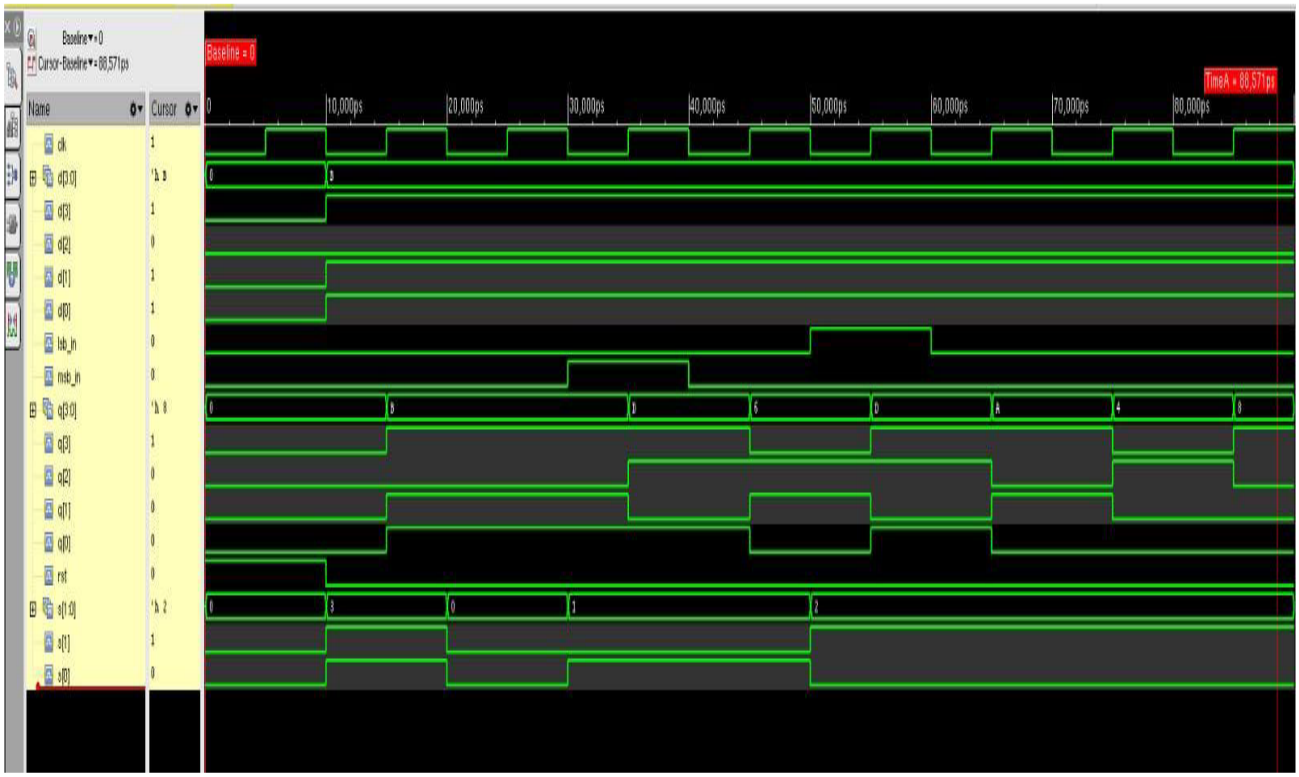
```
1  module shift_register (
2        input clk,
3        input reset,
4        input serial_in,
5        output reg [3:0] q
6  );
7     always @(posedge clk or posedge reset) begin
9        if (reset)
10           q <= 4'b0000;
11       else
12           q <= {q[2:0], serial_in};  // shift left and insert new bit
13    end
14 endmodule
15
```

Testbench:

```
15
16  module shift_register_tb;
17    reg clk, reset, serial_in;
18    wire [3:0] q;
18
20    shift_register uut (
21        .clk(clk),
22        .reset(reset),
23        .serial_in(serial_in),
24        .q(q)
25    );
26
27    always #5 clk = ~clk;
28
29    initial begin
30        clk = 0; reset = 1; serial_in = 0;
31        #10 reset = 0;
32        #10 serial_in = 1; #10 serial_in = 0; #10 serial_in = 1; #10 serial_in = 1;
33        #30 $finish;
34    end
35  endmodule
```

Line: 1 Col: 1 INS

Output Waveform:



Conclusion:

Thus, The 4-bit comparator was successfully designed and simulated using Verilog HDL. The circuit compares two 4-bit binary inputs and generates three outputs indicating whether the first input is greater than, less than, or equal to the second input. The simulation results verified the correct functioning of the comparator for different input combinations.

The 4-bit shift register was successfully designed and simulated using Verilog HDL. The circuit stores and shifts binary data by one bit position for every clock pulse. The simulation results verified the correct shifting of serial input data and the proper operation of the reset functionality.